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A Context-Aware Multimodal Retrieval-Augmented Generation Framework for Collaborative Robot Training

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ABSTRACT Collaborative robot training manuals combine procedural text with screens, wiring diagrams, and configuration images, limiting text-only retrieval-augmented generation (RAG) when learners require stage-specific evidence. This study proposes a context-aware multimodal RAG framework that separately indexes text chunks and visual materials from a collaborative robot manual. During preprocessing, bounding-box (BBox) distance filters spatially plausible text-image pairs, while SigLIP semantic similarity ranks the retained pairs. At runtime, the system combines text retrieval, image-only retrieval, page proximity, and text-image mapping metadata. It then estimates the training stage from the query and re-ranks candidates using stage-specific page ranges, section headings, and keywords. A local vector database and a 4-bit quantized large language model support offline deployment as a design objective. For evaluation, 70 queries were annotated with relevant text, a ground-truth image, and a training-stage label. Four groups were compared: keyword retrieval (G1), text-only RAG (G2), multimodal RAG (G3), and context-aware multimodal RAG (G4). Compared with G3, G4 increased Image Recall@5 from 71.4% to 77.1%, Image Recall@10 from 80.0% to 87.1%, and Image MRR from 0.572 to 0.620; Image Recall@1 increased from 45.7% to 48.6%. These results suggest that stage-aware re-ranking improves image-candidate ordering, although fine-grained image retrieval remains limited and response quality does not improve consistently. The contribution is therefore limited to retrieval-stage improvement on a single manual and internal query set; educational effects and actual performance on an 8 GB device remain unverified.

INDEX TERMS Collaborative robot training, multimodal RAG, retrieval-augmented generation, context-aware retrieval, engineering education, robot manual retrieval

I. INTRODUCTION

Collaborative robots are used in manufacturing, education, and research environments where humans and robots share a workspace. Prior work has emphasized the need for hands-on curricula that combine collaborative-robot concepts, safety, operation, and programming [1]. Because collaborative operation also requires safe and intuitive human-robot interaction [2], [3], learners must understand both procedural knowledge and the visual evidence presented in technical manuals.

The collaborative robot training manual analyzed in this study distributes procedural descriptions, teach pendant screens, wiring diagrams, and safety configuration images across multiple pages [30]. To identify information relevant to a particular training situation, learners must examine not only menu names and configuration items but also screen locations and associated images. Novice learners may also have difficulty connecting their current training stage to the corresponding section of the manual.

Retrieval-augmented generation (RAG) uses evidence retrieved from external documents to support language-model response generation [4]. Conventional text-only RAG primarily retrieves text chunks and may therefore provide a limited range of evidence for documents in which images, diagrams, and screen examples are necessary to understand a procedure. Because text and visual materials play complementary roles in collaborative robot training manuals, text retrieval alone may be insufficient to construct the required evidence.

To address this limitation, this study proposes a context-aware multimodal RAG framework for collaborative robot training.

The proposed system jointly retrieves text and visual materials from a manual and uses training-stage context to adjust their ranking. The central contribution is not an improvement in final answer generation itself, but the retrieval of more appropriate textual evidence and visual materials before answer generation.

The main contributions of this study are as follows.

1. A 70-query evaluation set was constructed from a collaborative robot training manual, with annotations for relevant text, ground-truth images, and training stages.
2. An experimental framework was designed to compare keyword retrieval, text-only RAG, multimodal RAG, and context-aware multimodal RAG using the same query set.
3. A multimodal retrieval architecture was implemented in which BBox distance serves as a spatial candidate filter and SigLIP semantic similarity serves as a candidate-ranking signal. The resulting text-image mapping is combined with text retrieval, image-only retrieval, and page proximity.
4. Query-based training-stage estimation and a stage-specific context map were applied to re-ranking, and their effects on image and diagram ranking and Both@k performance were evaluated.
5. Response quality was evaluated as a separate outcome, and representative improved and failed cases were examined to clarify the benefits and limitations of context-aware multimodal retrieval.

II. RELATED WORK

A. RETRIEVAL-AUGMENTED GENERATION

RAG supplements the parametric knowledge of an LLM with evidence retrieved from external documents [4]. Subsequent retrieval-augmented architectures have investigated the aggregation of multiple retrieved passages [5] and retrieval from large external memories during language modeling [6]. A typical RAG pipeline divides documents into chunks, embeds each chunk, stores the resulting representations, retrieves relevant evidence for a query, and supplies that evidence to a language model. Surveys of RAG systems consequently distinguish retrieval, augmentation, generation, and evaluation as related but separable components [11].

Retrieval methods used in RAG can broadly be divided into sparse and dense retrieval. BM25 is widely used as a representative keyword-based sparse retrieval baseline [8], whereas Dense Passage Retrieval (DPR) represents questions and passages using dense vectors [7]. Sentence-BERT further demonstrated efficient semantic similarity search using sentence embeddings [9]. In this study, G1 serves as the keyword-based baseline, while G2 implements multilingual dense retrieval using BGE-M3 embeddings [10].

RAG is useful for documents with well-defined domain knowledge, such as technical manuals, internal documentation, and educational materials. Text-only RAG, however, relies mainly on sentence- or paragraph-level text retrieval and is therefore limited when documents depend heavily on images, diagrams, tables, and screen captures. Collaborative robot training manuals are a representative example of this document type.

B. MULTIMODAL RETRIEVAL

Multimodal retrieval uses information from different modalities, such as text, images, diagrams, and audio, to identify relevant evidence. Vision-language models that map images and text into a shared embedding space enable text-to-image retrieval and image-text association. CLIP established a widely used contrastive image-text representation [12], and ALIGN demonstrated large-scale dual-encoder learning from noisy image-text pairs [13]. BLIP extended vision-language pretraining toward both understanding and generation tasks [14]. SigLIP replaces global softmax normalization with a pairwise sigmoid loss during image-text pretraining [15]. In this study, CLIP, ALIGN, and BLIP are discussed as related vision-language approaches, whereas SigLIP is the model used to compute text-image semantic similarity.

Multimodal RAG research has extended retrieval-augmented generation to external memories containing both text and images. MuRAG retrieves from a multimodal memory for question answering [16], REVEAL integrates multiple multimodal knowledge sources [17], and RA-CM3 retrieves and generates both text and images [18]. These methods establish the value of multimodal external evidence, but they do not directly address stage-aware retrieval from a compact technical training manual.

Training manuals frequently describe a single procedure using both text and images. A menu path may be explained in text, while the learner must also inspect a teach pendant or configuration screen. Document-understanding research has also shown that textual content and two-dimensional layout provide complementary information [19]. Independently returning text and image search results is therefore insufficient in the present setting. A retrieval system must consider how textual and visual materials are connected by page layout, surrounding context, and training stage.

C. CONTEXT-AWARE RETRIEVAL FOR TRAINING GUIDANCE

Queries in a practical training environment may be associated with particular task stages. Even when two queries contain similar terms, the required evidence and related image may differ depending on whether the learner is performing installation, configuring safety settings, or executing a program. Prior collaborative-robot education work has discussed the importance of laboratory practice and training modules involving safety, operation, and programming [1]. Research on industrial human-robot collaboration and ISO safety guidance also highlights the importance of safe, interpretable operation [2], [3]. Retrieval for training guidance should therefore consider the training stage and manual structure in addition to semantic query similarity.

This study estimates the training stage from the query and uses a context map containing stage-specific page ranges, section headings, and keywords to re-rank multimodal retrieval results.

III. PROPOSED FRAMEWORK

A. OVERALL ARCHITECTURE

The proposed system preprocesses textual and visual materials extracted from a collaborative robot training manual and indexes them for retrieval. When a user submits a training-related query, the system retrieves relevant text chunks and image candidates and generates a response using a selected local LLM.

The overall pipeline consists of the following steps.

1. Text and visual materials are extracted from the manual PDF.
2. Text is divided into chunks while retaining page numbers, section information, surrounding context, and text BBoxes.
3. Each extracted image is stored with its page number and image BBox.
4. Text-image pairs are collected from the same page and the following page. BBox distance restricts the spatial candidate set, and SigLIP semantic similarity ranks the remaining candidates to generate mapping metadata.
5. Text embeddings, the image index, and text-image mapping metadata are stored locally.
6. Text and image candidates are retrieved for the user query.
7. G3 combines text retrieval, image retrieval, page proximity, and precomputed SigLIP-based text-image mapping scores.
8. G4 estimates the training stage from the query and adjusts the candidate ranking using the corresponding context map.
9. The local LLM generates training guidance from the retrieved evidence.

B. TEXT PREPROCESSING AND INDEXING

Text extracted from the manual is divided into chunks while preserving procedural and semantic units. Each chunk stores the original text, page number, section-related metadata, and the BBox coordinates of the corresponding text region. A BBox represents the rectangular region occupied by text on a PDF page as `(x0, y0, x1, y1)`. BGE-M3 is used to generate multilingual text embeddings [10], and ChromaDB is used for persistent storage of embeddings, documents, and metadata [28].

In G2, the top-k text candidates are retrieved using the similarity between the query embedding and text-chunk embeddings. To make ranking deterministic for the fixed, compact corpus, the implementation loads the stored vectors and computes the squared Euclidean distance from the query to every text vector before sorting. The same exhaustive ranking procedure is applied to the stored image vectors used by G3 and G4. ChromaDB therefore serves as the persistent vector store, while the final experimental ranking bypasses approximate nearest-neighbor traversal. These candidates provide evidence for LLM-based answer generation.

C. IMAGE AND DIAGRAM PREPROCESSING AND RETRIEVAL

Images and diagrams extracted from the manual are managed using their filenames, page numbers, and image BBoxes. Because an isolated image may have limited semantic information, spatial and semantic associations between text chunks and images are computed during preprocessing. This use of spatial metadata is consistent with the broader observation that document layout carries information beyond text alone [19]. For each text chunk, images on the same page and the following page are collected as initial candidates. On the same page, the two-dimensional distance between the grouped text BBox and the center of the image BBox is calculated. The distance is adjusted to reflect the common manual layout in which an image appears below the corresponding text. A spatial penalty is applied to candidates on the following page.

BBox distance is not directly added to the final mapping score; it is used only as a candidate gate. Images with an adjusted distance below 300 points are retained as spatial candidates. To avoid discarding semantically related images with distant layouts, candidates are also rescued when the robust-normalized SigLIP similarity is at least 0.40. Candidate ranking and the `mapping_score` are determined using SigLIP image-text cosine similarity [15]. The cosine values are normalized to the range of 0 to 1 using robust boundaries derived from the preprocessing-pair distribution, and the same fixed boundaries are applied to the full dataset. BBox therefore removes spatially distant images, whereas SigLIP ranks the retained candidates by semantic relevance. These computations are performed once during offline preprocessing; the application uses only the stored mappings at runtime.

G3 combines the following signals for image retrieval:

- image-only retrieval score;
- proximity between retrieved text pages and image pages;
- precomputed SigLIP text-image mapping score after BBox candidate filtering; and
- baseline candidate-rank score.

This design retrieves not only textual evidence directly related to the query but also visual materials that are spatially and semantically connected to that evidence. Figure 1 presents the overall system architecture, including BBox filtering and SigLIP-based mapping. Figure 2 presents the G4 candidate expansion and re-ranking process that applies automatically inferred stage context to the G3 candidates.

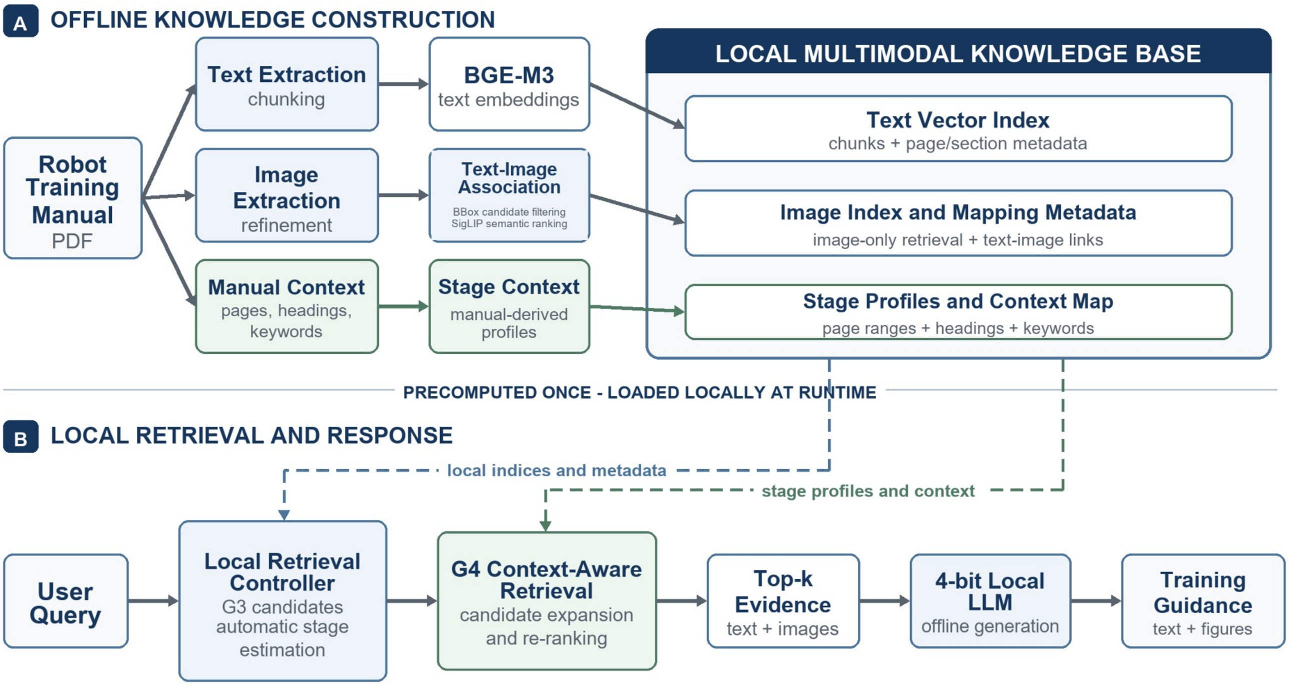

FIGURE 1. Overall architecture of the proposed context-aware multimodal RAG framework.

D. CONTEXT-AWARE RE-RANKING

G4 extends G3 by adding training-stage estimation and context-map-based re-ranking. For each training stage, the context map contains relevant page ranges, section headings, keywords, and mapping rationale. It was manually reviewed using the semantic structure of the manual and the meaning of each training stage. Ground-truth image filenames and ground-truth chunk identifiers were not used in the context map.

G4 performs re-ranking as follows.

1. BGE-M3 semantic similarity is computed between the query and each training-stage context profile to estimate the most relevant stage.
2. If the highest stage similarity is below 0.45 or the similarity margin between the first- and second-ranked stages is below 0.03, the system does not apply G4 re-ranking and instead falls back to the G3 ranking.
3. The page ranges and keywords associated with the estimated stage are loaded.
4. An additional score is assigned when a candidate falls within the relevant page range.
5. The score is adjusted when the surrounding text or section information matches the stage-specific keywords.
6. The G3 retrieval score and context score are combined to calculate the final ranking.

This procedure does not directly select candidates using ground-truth labels. Instead, it prioritizes candidates according to the manual structure and inferred training-stage context, and can therefore be interpreted as context-aware retrieval.

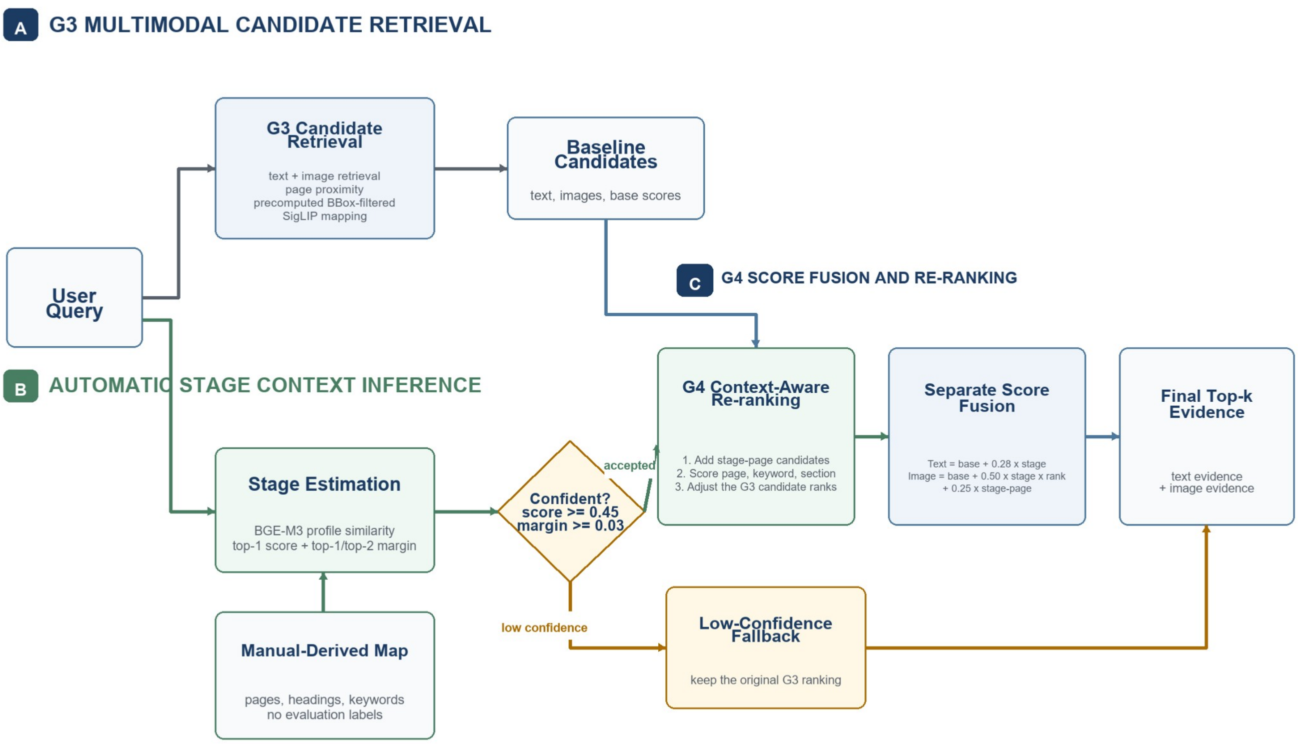

FIGURE 2. Context-aware candidate expansion and re-ranking process used in G4.

Algorithm 1 Context-Aware Multimodal Re-Ranking in G4

```
Input: query q; G3 text candidates T; G3 image candidates I; stage profiles S; context map C
Output: re-ranked Top-k text candidates T' and image candidates I'
1: Encode q using BGE-M3 and compute stage similarities a1 >= a2.
2: If a1 < tau_s or (a1 - a2) < tau_m, return the original G3 Top-k results.
3: Load page ranges, section terms, keywords, and stage weight for the inferred stage.
4: Expand I with valid image candidates from the inferred stage page range.
5: For each text candidate t:
6:     c_t = w_s(0.55 p_t + 0.30 k_t + 0.15 h_t).
7:     f_t = [1 - (rank_G3(t) - 1) / |T|] + beta_t c_t.
8: For each image candidate i:
9:     b_i = 0.25 v_i + 0.15 l_i + 0.50 n_i + 0.05 m_i + 0.05 d_i.
10:    c_i = w_s(0.50 p_i + 0.10 k_i + 0.40 h_i).
11:    Compute rho_i from the baseline rank window R.
12:    f_i = b_i + lambda_c c_i rho_i + lambda_p p_i.
13: Sort candidates by f_t and f_i in descending order and return Top-k.
Parameters: tau_s=0.45, tau_m=0.03, beta_t=0.28, lambda_c=0.50, lambda_p=0.25, R=120.
Symbols: p, k, h denote page, keyword, and section scores; v, l, n, m, d denote image-only, linked-text
rank, page proximity, stored SigLIP mapping, and diagram scores.
Leakage control: C contains no query IDs, ground-truth image filenames, or ground-truth chunk IDs.
```

The parameter values are linked to the limited grid search described in Section IV-D. They are fixed across all queries but were selected on the same internal 70-query pilot set used for final evaluation.

In Algorithm 1, `T` denotes the G3 text list expanded to at most 30 candidates when G4 is activated. `I` is the union of text-linked images, adjacent-page images, image-index candidates, and valid images within the inferred stage page range. `S` denotes the stage profiles constructed from stage names, section terms, content/action keywords, and manual evidence, whereas `C` stores the text and image page ranges and terms associated with each stage. `a_1` and `a_2` are the highest and second-highest cosine similarities between the query and stage profiles.

For a candidate `x`, `p_x`, `k_x`, and `h_x` denote stage-page, keyword, and section-heading match scores, respectively. The page score is 1.0 within the mapped range, 0.70 at a one-page distance, 0.45 at a two-page distance, and 0 otherwise. Keyword and section scores are normalized to the range [0,1] using matched terms in the candidate metadata. For an image candidate, `v_i`, `l_i`, `n_i`, `m_i`, and `d_i` denote image-only retrieval, linked-text rank, page proximity, stored SigLIP mapping after BBox candidate filtering, and diagram confidence, respectively. `b_x`, `c_x`, and `f_x` are the baseline, stage-context, and final scores. The attenuation factor `rho_i` prevents the context term from excessively promoting images that have weak baseline evidence.

If `a_1` is below `tau_s` or the difference between the two highest stage similarities is below `tau_m`, stage estimation is treated as insufficiently reliable and the method falls back to the G3 ranking. The thresholds and weights in Algorithm 1 are fixed for all queries, and their selection procedure is described in Section 4.4. To control experimental leakage, context map `C` contains no query IDs, ground-truth image filenames, or ground-truth chunk identifiers.

E. LOCAL LLM RESPONSE GENERATION AND RESOURCE-CONSTRAINED DESIGN

The system uses local LLMs to support deployment in resource-constrained environments. The implementation allows responses to be generated and compared using Qwen 2.5 [22], Gemma 2 [23], and Llama 3.1-family [24] models. Retrieved textual evidence and image-candidate information are supplied as context for generating training guidance.

For offline operation, the application uses Ollama-hosted local models and does not call an external LLM API at runtime. The selected models are 7B-9B Q4-quantized models, reflecting the broader use of low-bit quantization to reduce model memory and inference requirements [25], [26], [27]. These citations provide quantization background; the present system uses the Q4 model packages supplied through its local runtime rather than implementing QLoRA, GPTQ, or AWQ as a new contribution. BBox distance and SigLIP text-image similarity are computed during preprocessing, while runtime retrieval uses the stored indexes and mapping metadata. This configuration is intended to provide both textual evidence and related images without relying on a cloud-based model.

The quantitative evaluation focuses on the retrieval performance of G1-G4; offline execution speed and system performance in an 8 GB RAM environment were not measured. The resource-constrained configuration should therefore be interpreted as a design objective rather than a validated minimum hardware specification.

Response quality was evaluated separately from retrieval performance. A rule-based rubric evaluation was applied to all responses using correctness, specificity, training-stage relevance, safety, and comprehensibility criteria. Section 6 describes the evaluation criteria and procedure.

IV. EXPERIMENTAL DESIGN

A. DATASET

The experimental dataset was constructed from the Doosan Robotics DART-Platform manual [30]. The manual and its visual materials were used in this study with permission from Doosan Robotics, and the source manual is cited in [30]. A total of 70 queries were created, and each query was annotated with relevant text, a ground-truth image, and a training-stage label. The queries cover procedures frequently encountered in collaborative robot training, including installation, power supply, teach pendant operation, direct teaching, coordinate configuration, safety configuration, I/O wiring, and system management.

The query set was designed to emphasize questions for which a ground-truth image could be clearly identified, enabling quantitative evaluation of multimodal image retrieval.

B. COMPARISON GROUPS

The experiment compared the following four groups.

TABLE I
COMPARISON GROUPS

Group	Name	Description
G1	Keyword Search	Retrieves text chunks based on keyword overlap with the query
G2	Text-only RAG	Retrieves text chunks using BGE-M3 text embeddings
G3	Multimodal RAG	Combines text retrieval, image retrieval, page proximity, and BBox-filtered SigLIP text-image mapping
G4	Context-aware Multimodal RAG	Estimates the training stage from the query and applies context-map-based re-ranking to G3 candidates

G1 and G2 serve as baselines for text retrieval. G3 adds image and diagram retrieval, while G4 is the proposed method that incorporates training-stage information to re-rank multimodal candidates.

C. EVALUATION METRICS

Retrieval performance was evaluated using Recall@k and mean reciprocal rank (MRR). MRR assigns each query the reciprocal rank of its first relevant result and averages this value across queries, following established ranked question-answering evaluation practice [29].

Text Recall@k measures the proportion of queries for which the top-k text candidates contain the relevant text or equivalent information. Image Recall@k measures the proportion of queries for which the top-k image candidates contain the ground-truth image. MRR evaluates how highly the first relevant candidate is ranked. From a multimodal perspective, Both@k measures whether both the relevant text and ground-truth image are present within their respective top-k results.

Text retrieval was evaluated using a relaxed relevance criterion. A retrieved text candidate was considered relevant if it was located on the same page or in the same section as the

reference evidence, contained essential keywords from the reference answer, or conveyed semantically equivalent procedural or configuration information. This criterion was adopted because procedural information in the manual may be distributed across adjacent chunks or multiple passages on the same page rather than appearing in a single fixed sentence.
Image retrieval was evaluated using strict filename matching. An image was considered correct only when the exact ground-truth filename appeared among the top-k image candidates. A visually similar image from the same page was counted as incorrect if its filename differed. Consequently, Text Recall@k and Text MRR in this paper represent relaxed text retrieval performance, whereas Image Recall@k and Image MRR represent strict image retrieval performance. Both@k requires a relaxed text hit and a strict image hit simultaneously.

TABLE II
RELEVANCE CRITERIA FOR RETRIEVAL EVALUATION

Metric	Relevance Criterion
Text Recall@k / Text MRR	Relaxed: at least one of page/section agreement, essential-keyword inclusion, or semantic equivalence
Image Recall@k / Image MRR	Strict: the exact ground-truth image filename must appear in the top-k results
Both@k	A relaxed text hit and a strict image hit must both be satisfied

D. RETRIEVAL SCORE COMPOSITION AND WEIGHT SELECTION

The baseline G3 image score is a weighted sum of image-only retrieval (0.25), text-candidate rank (0.15), page proximity between the retrieved text and image (0.50), precomputed SigLIP mapping (0.05), and diagram confidence (0.05). BBox distance is not included in this weighted sum and is used only as a preprocessing candidate filter. The G4 image context score is composed of page (0.50), keyword (0.10), and section (0.40) scores. The final G4 image score additionally includes the context score with a weight of 0.50 and a stage-page prior with a weight of 0.25. A rank factor that decreases with the baseline G3 rank is applied to the context term. A factor of 0.35 is used for candidates outside the top 120 baseline ranks when they fall within the inferred stage page range. For text re-ranking, 0.28 times the stage-context score is added to the baseline retrieval-rank score.
Using the notation in Algorithm 1, the minimum stage similarity `tau_s` is 0.45, the minimum margin `tau_m` is 0.03, the text-context coefficient `beta_t` is 0.28, the image-context coefficient `lambda_c` is 0.50, the stage-page prior coefficient `lambda_p` is 0.25, and the baseline-rank attenuation window `R` is 120. These values are fixed implementation parameters applied uniformly to all evaluated queries rather than query-specific settings.
The weights were compared through a limited grid search using a candidate-level feature cache generated from the 70 queries. Strict Image MRR was used as the primary selection criterion, with Recall@5 and Recall@10 as secondary metrics. Five-fold cross-validation was also used to examine fold-level stability. The configuration that used BBox only as a candidate filter and SigLIP to rank the mapped candidates achieved higher average performance than configurations that directly added BBox proximity to the mapping score. However, because weight selection and final evaluation used the same 70-query internal pilot dataset, the reported values should not be interpreted as evidence of generalization to an independent external test set. A separate hold-out set or evaluation using an additional manual is required for the final paper.
The following compact comparison fixes the remaining G3 retrieval components and varies only the text-image mapping design. It is presented as a supporting analysis for separating the roles of BBox and SigLIP rather than as an independent evaluation of the two components.

TABLE III
COMPARISON OF BBOX- AND SIGLIP-BASED TEXT-IMAGE MAPPING

Text-Image Mapping Configuration	Image R@1	Image R@5	Image R@10	Image MRR
BBox-based mapping score	37.1%	68.6%	80.0%	0.518
BBox candidate filter + SigLIP ranking (final)	45.7%	71.4%	80.0%	0.572

E. EXPERIMENTAL ENVIRONMENT

The experiments were performed on a local laptop. Although the study targets an offline RAG system that can operate under constrained resources, the current quantitative experiments were conducted on the development and evaluation machine described below. The approximately 8 GB RAM target is therefore presented as a design and model-selection objective. Runtime, memory consumption, and concurrent stability on an actual system with 8 GB or less remain part of future reproducibility validation. The retrieval and response-quality tables reported in this paper compare groups under the same development environment and are not benchmarks measured on an 8 GB RAM device.

TABLE IV
EXPERIMENTAL ENVIRONMENT

Item	Specification
Operating system	Windows
Device	HP OMEN Gaming Laptop 16-am0xxx
CPU	Intel Core Ultra 7 255H, 16 cores / 16 logical processors
RAM	Approximately 24 GB
GPU	NVIDIA GeForce RTX 5060 Laptop GPU and Intel Graphics
Python	3.12.10 (.venv)
Vector database	ChromaDB 1.5.9
Application framework	Streamlit 1.58.0
Embedding / ML stack	sentence-transformers 5.6.0, transformers 5.12.1, torch 2.12.1
Data processing	pandas 3.0.3, PyMuPDF 1.27.2.3, Pillow 12.2.0

The local LLMs were configured to run through Ollama. The primary models used by the comparison applications are listed below.

TABLE V
LOCAL LANGUAGE MODELS

Model	Local Model ID	Application File	Local Storage Size
Qwen 2.5 7B Q4	qwen2.5:7b	src/app_qwen.py	Approximately 4.36 GB
Gemma 2 9B Q4	gemma2:9b	src/app_gemma.py	Approximately 5.07 GB
Llama 3.1 8B Q4	llama3.1:8b	src/app_llama.py	Approximately 4.58 GB

Other locally stored models were excluded from the comparison. The primary comparison was limited to Qwen 2.5 7B, Gemma 2 9B, and Llama 3.1 8B.

V. EXPERIMENTAL RESULTS

A. OVERALL RETRIEVAL PERFORMANCE

Table VI summarizes the final text, image, and joint retrieval performance of G1, G2, G3, and G4 under the current implementation.

TABLE VI
RETRIEVAL PERFORMANCE OF G1-G4

Group	Text R@1	Text R@5	Text R@10	Text MRR	Image R@1	Image R@5	Image R@10	Image MRR	Both@5	Both@10
G1 Keyword Search	75.7%	95.7%	98.6%	0.837	-	-	-	-	-	-
G2 Text-only RAG	81.4%	95.7%	100.0%	0.879	-	-	-	-	-	-
G3 Multimodal RAG	81.4%	95.7%	100.0%	0.879	45.7%	71.4%	80.0%	0.572	71.4%	80.0%
G4 Context-aware Multimodal RAG	85.7%	95.7%	100.0%	0.903	48.6%	77.1%	87.1%	0.620	77.1%	87.1%

Despite being a simple keyword-based method, G1 achieved a Text Recall@5 of 95.7%. This can be attributed to the explicit technical terms in collaborative robot manual queries, including menu names, configuration values, and function names. G2 nevertheless outperformed G1 in Text Recall@1 and Text MRR, indicating that embedding-based text retrieval was more effective at placing the relevant text at a higher rank.
G2 and G3 produced identical text retrieval results because both groups use the same BGE-M3 text embedding model, ChromaDB text collection, text chunks, and top-k text retrieval procedure. G3 does not modify the G2 text retriever; instead, it extends the G2 results with image-only retrieval, page proximity, and text-image mapping scores. The identical Text Recall@k and Text MRR values are therefore an intended consequence of the experimental design rather than an implementation error. G3 is evaluated for its ability to retrieve relevant images and diagrams in addition to text, not for improving text retrieval.
G3 achieved an Image Recall@5 of 71.4% and an Image Recall@10 of 80.0%, while Image Recall@1 and Image MRR were 45.7% and 0.572, respectively. Using BBox only as a candidate filter and combining SigLIP mapping with page proximity improved top-ranked retrieval quality relative to the earlier G3 pilot configuration. Nevertheless, fewer than half of the queries placed the ground-truth image at rank one, indicating substantial room for improvement.
G4 produced numerically higher image-retrieval metrics than G3. Image Recall@5 increased from 71.4% to 77.1%, Image Recall@10 increased from 80.0% to 87.1%, and Image MRR increased from 0.572 to 0.620. Both@5 and Both@10 likewise increased from 71.4% to 77.1% and from 80.0% to 87.1%, respectively. Image Recall@1 increased by 2.9 percentage points, from 45.7% to 48.6%. These results indicate that G4 moved some relevant images into the top-five or top-ten range and improved their average rank rather than consistently placing the ground-truth image at rank one. Because no statistical significance test was included, these numerical increases should not be interpreted as evidence of general superiority.

B. CASES IMPROVED BY G4

Aggregate metrics do not fully explain how G4 changes the candidate ranking. This subsection therefore examines queries for which the ground-truth image rank improved relative to G3. These cases demonstrate how query-based stage estimation, page ranges, section headings, and keyword context can move relevant images upward.
For Q31, which asks about the primary purpose of the USB port on the teach pendant, the ground-truth image `page_103_img_0_0.jpeg` moved from rank eight in G3 to rank three in G4. The query was associated with the `teach pendant/USB data management` stage, and the context included terms related to USB ports, the pendant, and data import/export.
For Q23, which asks where to check the system time, the ground-truth image `page_167_img_3_0.jpeg` was outside the top ten in G3 but moved to rank four in G4. The query was associated with the `UI/system information inspection` stage, and context related to system information, time, and screen areas contributed to re-ranking.
For Q02, which asks about the grounding conditions required to supply power to the controller, the ground-truth image `page_403_img_0_0.jpeg` moved from rank two in G3 to rank one in G4. The query was associated with the `safety/power/grounding` stage, and context involving grounding, power supply, the controller, and safety was applied.
These cases show that G4 can use training-stage information to improve the rank of the correct image for some queries. They do not, however, imply consistent top-one retrieval across the full query set.

C. FAILURE CASES OF G4

Reporting only improved cases would overstate the effectiveness of G4. The following failures demonstrate that page- and section-level context does not fully resolve ambiguity among similar images, compensate for missing fine-grained screen cues, or eliminate query-based stage-estimation errors.
Q10 asks for the menu path used to inspect the current angle of each robot joint on the teach pendant. Although G4 considered Status-related pages and keywords, it failed to retrieve the ground-truth image `page_332_img_0_0.jpeg` within the top ten. Terms such as `Status`, `I/O Overview`, and `joint angle` did not contribute strongly enough during candidate collection.
Q28 asks which cable-coupling component is required to strengthen the waterproof rating of the robot cube module cable. G4 moved several images near page 405 upward, but the exact ground-truth image `page_405_img_0_0.jpeg`

remained outside the top ten. This case reveals insufficient image-level discrimination among multiple images on the same page.

Q15 asks for the menu path used to export historical system error logs. The stage estimator favored `UI/system information inspection` rather than `system management/logs`, and the ground-truth image `page_340_img_0_0.jpeg` did not appear within the top ten. This case demonstrates that G4 performance depends on the accuracy of stage estimation.

These failures also clarify the role of the BBox candidate filter. BBox is useful for eliminating spatially distant images, but it does not distinguish similar images that pass the filter on the same page. The current ranking therefore depends on SigLIP semantic similarity and page- and section-level context, which may not capture small differences within a screen or the ordering of multiple images. Image captions, local text surrounding each image BBox, image-order information, and more fine-grained stage labels are possible directions for addressing this limitation.

VI. RESPONSE QUALITY EVALUATION

Retrieval performance and local-LLM response quality were evaluated as distinct outcomes. Responses were generated under G2, G3, and G4 using Qwen, Gemma, and Llama, yielding 630 responses across 70 queries.

Response quality was assessed separately because RAG evaluation encompasses context relevance, answer faithfulness, and answer relevance in addition to retrieval accuracy [20], [21]. The following five criteria were used.

1. Correctness: Does the response agree with the essential content of the reference answer?
2. Specificity: Does the response provide sufficiently specific menu paths, button names, configuration values, or procedures?
3. Training-stage relevance: Is the guidance appropriate for the training stage represented by the query?
4. Safety: Does the response avoid unsafe or incorrect robot-operation instructions?
5. Comprehensibility: Is the explanation understandable to a novice learner?

Each criterion was scored from 1 to 5, and the average was used to compare models and retrieval groups. All 630 responses were processed using a rule-based rubric based on reference-keyword agreement, semantic similarity, and safety and readability rules.

TABLE VII
RESPONSE-QUALITY RESULTS

Group	Model	N	Mean Score	O	Partial	X
G2 Text-only RAG	Qwen	70	4.05	49	13	8
G2 Text-only RAG	Gemma	70	3.59	27	31	12
G2 Text-only RAG	Llama	70	3.62	24	33	13
G3 Multimodal RAG	Qwen	70	4.01	47	15	8
G3 Multimodal RAG	Gemma	70	3.56	27	29	14
G3 Multimodal RAG	Llama	70	3.60	28	27	15
G4 Context-aware Multimodal RAG	Qwen	70	3.99	47	14	9
G4 Context-aware Multimodal RAG	Gemma	70	3.59	27	31	12
G4 Context-aware Multimodal RAG	Llama	70	3.62	31	26	13

Qwen achieved higher mean scores than the other two models in all three retrieval groups. Unlike retrieval performance, however, response quality did not improve consistently under G4. Qwen achieved mean scores of 4.05 in G2, 4.01 in G3, and 3.99 in G4, indicating a small decrease under G4. Gemma increased slightly from 3.56 in G3 to 3.59 in G4, while Llama increased from 3.60 to 3.62.

These results show that a numerical increase in retrieval performance did not directly translate into improved response quality. Generated responses were jointly affected by changes in the retrieved evidence, query-based stage estimation, and the generation characteristics of each local LLM. The contribution of this study should therefore be interpreted as a limited improvement at the retrieval stage. This rule-based response-quality evaluation does not substitute for judgments by users or domain experts and is not used as evidence of educational effectiveness.

VII. DISCUSSION

G1 already achieved a high Text Recall@5, and G2 and G3 reached a Text Recall@10 of 100.0%. In contrast, G3 achieved an Image Recall@1 of 45.7%. However, text retrieval used a relaxed relevance criterion, whereas image retrieval used strict filename matching. The absolute values of the two modalities therefore cannot be directly compared to conclude that image retrieval is inherently more difficult. Nevertheless, under the strict criterion, the ground-truth image was ranked first for fewer than half of the queries, indicating that image ranking remains a major target for improvement.

G4 achieved numerically higher Image Recall@5, Image Recall@10, and Image MRR than G3. In several cases with clearly defined stage-related page ranges and keywords, the ground-truth image moved upward in the ranking. Nevertheless, Image Recall@1 remained at 48.6%, and neither statistical significance nor generalization to external data was evaluated. G4 should therefore be interpreted as an internal pilot method for adjusting a candidate ranking rather than as a method that reliably selects a single correct image.

G4 incorporates page- and section-level context, but it could not reliably distinguish multiple images on the same page or similar configuration screens within the same section. Because BBox is a preprocessing candidate gate, it does not directly discriminate among similar images that pass the filter. Local text around each BBox, image-level captions, image-order information, and more fine-grained training-stage classification are possible directions for improving image-level discrimination.

In the response-quality evaluation, G4-Qwen achieved a mean score of 3.99, which was higher than the other two models under G4 but lower than Qwen under G2 and G3. This result provides limited evidence that a quantized 7B-class local model can generate training guidance, but improvements in retrieval metrics did not consistently translate into better generated responses. The contribution of G4 is therefore centered on evidence retrieval and image-candidate ranking rather than generation quality.

VIII. CONCLUSION

This study proposed a context-aware multimodal RAG framework for collaborative robot training and evaluated retrieval performance using 70 queries derived from a collaborative robot training manual. Text-only RAG achieved high Recall@k for manual-text retrieval, and G3 combined the text results with image and diagram retrieval signals.

G4 estimated the training stage from the query and re-ranked G3 candidates using a stage-specific context map. Image Recall@5 increased from 71.4% to 77.1%, Image Recall@10 increased from 80.0% to 87.1%, and Image MRR increased from 0.572 to 0.620. However, Image Recall@1 remained below 50% at 48.6%, and weight selection and final evaluation used the same dataset. The results should therefore be interpreted as an improvement in relevant image-candidate ranking on the internal pilot dataset rather than as a resolution of the image-retrieval problem.

The results demonstrate the applicability of stage-context-aware multimodal re-ranking to collaborative robot training manuals that jointly use text, images, diagrams, and procedural information. However, response quality did not improve consistently, and the framework was not evaluated on independent data or on an actual system with approximately 8 GB of RAM. Accordingly, the present contribution is limited to retrieval-stage improvements under the reported experimental setting. Subsequent work will examine fine-grained image discrimination and prompt and evidence composition, followed by validation on separate data and evaluation with learners or domain experts before claims concerning educational effectiveness are made.

REFERENCES

[1] A. Djuric, J. L. Rickli, V. M. Jovanovic, and D. Foster, "Hands-on learning environment and educational curriculum on collaborative robotics," in Proc. 2017 ASEE Annu. Conf. Expo., Columbus, OH, USA, 2017, pp. 1-15.
[2] V. Villani, F. Pini, F. Leali, and C. Secchi, "Survey on human-robot collaboration in industrial settings: Safety, intuitive interfaces and applications," Mechatronics, vol. 55, pp. 248-266, Nov. 2018, doi: 10.1016/j.mechatronics.2018.02.009.
[3] Robots and Robotic Devices--Collaborative Robots, ISO/TS 15066:2016, International Organization for Standardization, Geneva, Switzerland, 2016.
[4] P. Lewis et al., "Retrieval-augmented generation for knowledge-intensive NLP tasks," in Advances in Neural Information Processing Systems, vol. 33, 2020, pp. 9459-9474.
[5] G. Izacard and E. Grave, "Leveraging passage retrieval with generative models for open domain question answering," in Proc. 16th Conf. Eur. Chapter Assoc. Comput. Linguistics, 2021, pp. 874-880, doi: 10.18653/v1/2021.eacl-main.74.
[6] S. Borgeaud et al., "Improving language models by retrieving from trillions of tokens," in Proc. 39th Int. Conf. Mach. Learn., vol. 162, 2022, pp. 2206-2240.
[7] V. Karpukhin et al., "Dense passage retrieval for open-domain question answering," in Proc. 2020 Conf. Empirical Methods Natural Language Process., 2020, pp. 6769-6781, doi: 10.18653/v1/2020.emnlp-main.550.
[8] S. Robertson and H. Zaragoza, "The probabilistic relevance framework: BM25 and beyond," Found. Trends Inf. Retr., vol. 3, no. 4, pp. 333-389, 2009, doi: 10.1561/1500000019.
[9] N. Reimers and I. Gurevych, "Sentence-BERT: Sentence embeddings using Siamese BERT-networks," in Proc. 2019 Conf. Empirical Methods Natural Language Process. and 9th Int. Joint Conf. Natural Language Process., 2019, pp. 3982-3992, doi: 10.18653/v1/D19-1410.
[10] J. Chen, S. Xiao, P. Zhang, K. Luo, D. Lian, and Z. Liu, "M3-Embedding: Multi-linguality, multi-functionality, multi-granularity text embeddings through self-knowledge distillation," in Findings Assoc. Comput. Linguistics: ACL 2024, 2024, pp. 2318-2335, doi: 10.18653/v1/2024.findings-acl.137.
[11] Y. Gao et al., "Retrieval-augmented generation for large language models: A survey," arXiv:2312.10997, 2023.
[12] A. Radford et al., "Learning transferable visual models from natural language supervision," in Proc. 38th Int. Conf. Mach. Learn., vol. 139, 2021, pp. 8748-8763.
[13] C. Jia et al., "Scaling up visual and vision-language representation learning with noisy text supervision," in Proc. 38th Int. Conf. Mach. Learn., vol. 139, 2021, pp. 4904-4916.
[14] J. Li, D. Li, C. Xiong, and S. Hoi, "BLIP: Bootstrapping language-image pre-training for unified vision-language understanding and generation," in Proc. 39th Int. Conf. Mach. Learn., vol. 162, 2022, pp. 12888-12900.
[15] X. Zhai, B. Mustafa, A. Kolesnikov, and L. Beyer, "Sigmoid loss for language image pre-training," in Proc. IEEE/CVF Int. Conf. Comput. Vis., 2023, pp. 11975-11986.
[16] W. Chen, H. Hu, X. Chen, P. Verga, and W. W. Cohen, "MuRAG: Multimodal retrieval-augmented generator for open question answering over images and text," in Proc. 2022 Conf. Empirical Methods Natural Language Process., 2022, pp. 5558-5570, doi: 10.18653/v1/2022.emnlp-main.375.
[17] Z. Hu et al., "REVEAL: Retrieval-augmented visual-language pre-training with multi-source multimodal knowledge memory," in Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit., 2023, pp. 23369-23379.
[18] M. Yasunaga et al., "Retrieval-augmented multimodal language modeling," in Proc. 40th Int. Conf. Mach. Learn., vol. 202, 2023, pp. 39755-39769.
[19] Y. Xu, M. Li, L. Cui, S. Huang, F. Wei, and M. Zhou, "LayoutLM: Pre-training of text and layout for document image understanding," in

Proc. 26th ACM SIGKDD Int. Conf. Knowl. Discovery Data Mining, 2020, pp. 1192-1200, doi: 10.1145/3394486.3403172.
[20] S. Es, J. James, L. Espinosa-Anke, and S. Schockaert, "RAGAs: Automated evaluation of retrieval augmented generation," in Proc. 18th Conf. Eur. Chapter Assoc. Comput. Linguistics: Syst. Demonstrations, 2024, pp. 150-158, doi: 10.18653/v1/2024.eacl-demo.16.
[21] J. Saad-Falcon, O. Khattab, C. Potts, and M. Zaharia, "ARES: An automated evaluation framework for retrieval-augmented generation systems," in Proc. 2024 Conf. North Amer. Chapter Assoc. Comput. Linguistics: Human Language Technol., 2024, pp. 338-354, doi: 10.18653/v1/2024.naacl-long.20.
[22] Qwen Team, "Qwen2.5 technical report," arXiv:2412.15115, 2024.
[23] Gemma Team et al., "Gemma 2: Improving open language models at a practical size," arXiv:2408.00118, 2024.
[24] A. Grattafiori et al., "The Llama 3 herd of models," arXiv:2407.21783, 2024.
[25] T. Dettmers, A. Pagnoni, A. Holtzman, and L. Zettlemoyer, "QLoRA: Efficient finetuning of quantized LLMs," in Advances in Neural Information Processing Systems, vol. 36, 2023.
[26] E. Frantar, S. Ashkboos, T. Hoefler, and D. Alistarh, "GPTQ: Accurate post-training quantization for generative pre-trained transformers," in Proc. 11th Int. Conf. Learn. Representations, 2023.
[27] J. Lin et al., "AWQ: Activation-aware weight quantization for on-device LLM compression and acceleration," Proc. Mach. Learn. Syst., vol. 6, pp. 87-100, 2024.
[28] Chroma, "Chroma documentation: Introduction." Accessed: Jul. 25, 2026. [Online]. Available: https://docs.trychroma.com/docs/overview/introduction
[29] E. M. Voorhees and D. M. Tice, "The TREC-8 question answering track evaluation," in Proc. 8th Text REtrieval Conf., NIST Special Publication 500-246, 2000, pp. 83-105.
[30] Doosan Robotics, "DART-Platform manual," User Manual, ver. 3.6.0. Accessed: Jul. 25, 2026. [Online]. Available: https://manual.doosanrobotics.com/en/user-manual/3.6.0/1-m-h-series/part-6-dart-platform-manual

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